Magpie — Sprint 20 spec: profit intelligence v1

Audience: Codex. **Author:** Claude. **Status: ready to implement. Locks:** D-131 (sequencing — step 4; the "cash-lock" view is the stale/velocity feature per D-131), D-129 (no AI numbers), D-121. **Facts:** F-09 (no scraping), F-19 (2026 fee model), F-20 (no new eBay API usage). **Builds on:** Sprint 18 (fee engine + all-in cost framing), Sprint 12 (dashboard/analytics patterns incl. the negative-duration guard), Sprint 19 (rule-stated honesty labels), live at `bf2c17d`.

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1. Goal

Turn Magpie's owned sales + inventory data into **decision-driving profit intelligence**: what did each sale *really* make (2026-correct fees, all-in costs), **how fast** money comes back (**profit per day held** — the one velocity signal), **where cash is trapped** (the cash-lock view + stale nudge), **what to buy more of** (one honest ranking), and a **financial-year ledger export** for the accountant — timely, since **FY2025-26 closed 30 June**. Everything computed live from owned data; nothing forecast, nothing fabricated.

2. Scope

IN:

- **Realised P&L ledger** (§4) · **profit-per-day-held** (§5) · **cash-lock view + stale nudge** (§6) · **one "buy more of this" ranking** (§7) · **AU FY ledger export** (§8) · a dedicated mobile-first **Profit page** (+ optional new KPI-tile allow-list entries, §9).

OUT (explicitly):

- ❌ **Forecasting or predicted values of any kind** — realised + derived-from-owned-data metrics only. ❌ AI anywhere in these paths (D-129). ❌ Scraping / new eBay API calls (F-09 / F-20 gate). ❌ **The source/supplier dimension** — deliberately deferred to **Sprint 21** where source tagging lands (rank by category + channel only here). ❌ GST/BAS calculations or any tax computation — the export is **records, not tax advice**. ❌ Lot cost-allocation (Sprint 21) and multi-channel packs (22).

3. Data model (additive-minimal; inspect first)

- P&L derives from existing `SaleRecord` + `Item` cost fields + the Sprint-18 fee engine **— expect no new models**.
- **Listing timestamp for staleness:** use the best existing signal (the publish/listing record from Phase 4) if one exists; **only if absent**, add a nullable `listed_at` to the listing/draft or item (additive), set on publish and manually editable (for items listed outside Magpie).
- All aggregates computed live (Sprint-12 pattern); no warehouse tables.

4. Realised P&L ledger

Per **sold** item/sale: **revenue (seller_receives basis) − fees(seller mode, Sprint-18 engine) − all-in costs (acquisition + refurb + packaging + postage/label + other direct costs) = realised profit**, plus all-in ROI and days held. A ledger view (rows per sale, drill-through to item) + aggregates by **category** and **channel**. Losses display honestly as losses (the deliberate loss-making test sale is a natural test case — it must show negative, not be smoothed).

5. Velocity: profit per day held (the one signal)

- `days_held = max(1, sale_date − acquisition_date)` **— reuse the Sprint-12 negative/zero-duration guard**; never divide by less than 1.
- **Headline metric: profit ÷ days held** (per sale; median per group). Show **annualised all-in ROI as secondary context only** — profit-per-day is the decision number. Tooltip states both formulas.

6. Cash-lock view + stale nudge

- **Cash locked = all-in cost basis of unsold stock**, bucketed: **unlisted / listed-fresh / listed-stale**, with totals and per-item drill-through. Stale threshold **configurable, default 90 days** since `listed_at`.
- **Stale nudge:** stale items carry a visible "listed N days — reprice or relist?" flag (a flag/list, **not** any automated action).
- Items with no `listed_at` and no publish record appear in **unlisted** (honest default), with an inline hint to set the date if they're actually listed elsewhere.

7. "Buy more of this" — one ranking, honestly thin

- **One view:** groups by **category + channel**, ranked by **median profit-per-day-held**, showing n, median profit, median days held, newest sale date.
- **Rule-stated threshold (Sprint-19 pattern):** only groups with **n ≥ 3** are ranked; smaller groups appear unranked as "insufficient data (n = X)". The rule is visible in a tooltip. **No invented confidence.** With only ~4 sales today, this view will read almost entirely thin — **that is correct behaviour**, not a bug; it fills as the business trades.

8. AU financial-year ledger export

- FY selector (**Jul–Jun**; current + past FYs present in data). Per-FY **summary** (revenue on seller_receives basis, total fees, total costs, realised profit, sale count) + **CSV export** of per-sale rows: dates (acquired, listed, sold), item ID/title, category, channel, seller mode, revenue, fee breakdown, cost components, realised profit, days held.
- **Label prominently: "Sale records for your accountant — not tax advice."** No GST/BAS logic, no tax fields invented. FY boundary handling tested (a 30 June sale vs a 1 July sale land in different FYs).

9. Design

A dedicated **Profit page**, mobile-first, Sprint-16 bright high-contrast tokens; ledger tables with tabular figures; cash-lock buckets as clear stacked totals; the ranking as a simple list. Optionally add **"profit/day (median)"** and **"cash locked"** to the dashboard's KPI-tile **allow-list** (extending the existing D-104 mechanism — allow-list validation included). No overflow at 390 px.

10. Acceptance criteria (closure gate)

- **P&L correctness:** per-sale profit matches the Sprint-18 fee engine + all-in costs across seller modes (tests); **the loss-making test sale shows a negative profit** (test).
- **Velocity:** days-held guard (never < 1; negative-duration edge covered); profit-per-day + annualised context correct (tests).
- **Cash-lock:** bucketing correct incl. the no-`listed_at` → unlisted default; stale threshold configurable; nudge appears at threshold (tests).
- **Ranking:** n ≥ 3 rule enforced + visible; below-threshold groups shown unranked

(tests).

- **FY export:** boundary tests (30 Jun / 1 Jul); CSV golden-file test (columns + a known row); the not-tax-advice label present.
- **Honesty:** thin states throughout read as thin; **no forecasts anywhere** (assert no predictive fields in responses).
- **No new network calls** (assert; F-09/F-20). **No AI in these paths.**
- **Process:** suites + typecheck/build/collectstatic green; **dual-lane Validation green**; additive migration only if `listed_at` is needed (then **pre-migration encrypted backup + restore spot-check**); Administrator restart (known service permissions); **phone screenshots headline** (Profit page, cash-lock, FY export). Evidence as `SPRINT20_EVIDENCE.md`.

11. For Codex

Reasoning: High. Everything derives from **owned data through the Sprint-18 fee engine** — one profit definition everywhere (no parallel maths; if the calculator/ledger ever need shared helpers, extract, don't duplicate — the parity-fixture lesson). **Hard lines:** no AI numbers (D-129), no forecasting, no scraping / new eBay API (F-09/F-20), losses shown as losses, thin shown as thin. `days_held` guard per Sprint 12. Additive-minimal schema (inspect before adding `listed_at`). **Never echo `.env` values** — the credential rotation is still in progress. Evidence as `SPRINT20_EVIDENCE.md`. Next per D-131: Sprint 21 = lot purchase manager + source ROI.